Seldon Manifolds — Decision Making

Narrative Continuation, Target Shapes, Surprise, and Meta-analysis

Matthew Parslow
Independent Researcher

Draft — draft-2026-09-06.1
6 September 2026 • document revision 1

DRAFT — draft-2026-09-06.1r1 (6 September 2026). Circulated for comment, not as a finished result. This is the decision-making companion to *Seldon Manifolds — Theory* (referred to below as *Seldon*) and inherits its definitions, together with those of *The Law of Graphic Equalisation — Theory*, the momentum paper, and *Narrativium*. *Seldon* defines a represented geometry of reachable futures; this paper specifies a proposed decision grammar over that geometry. It does not claim that human, organisational, or artificial agents instantiate that grammar. Its criterion is comparative operational usefulness under a stated scope, baseline, and inference budget, not abstract truth. Sections marked as operational commitments are intended points of attack.

Abstract

A Seldon manifold represents the future states or histories distinguishably reachable in a frame under represented counterfactual interventions and boundary conditions. It does not by itself decide among them. This paper supplies a proposed decision grammar. Decision is evaluated only over the frame's represented landscape. The standing Narrative is the continuation requiring zero *additional decision inference*; implementation may be costly without changing it. A decision is reopened either by represented surprise or by a costly, frame-triggered meta-analysis. Direct revision through surprise requires a resolution already available within aperture and coherent in the current frame; otherwise the surprise may motivate meta-analysis but cannot directly replace Narrative.

Targets are represented acceptable shapes in the manifold rather than points with error bars. Narrativium supplies a feasibility gate, Pareto comparison does not supply selection, and no future self can authorise the transition that first installs it. The grammar is agent-neutral: individual, business, state, and collective differ through frame contents and coupling, not decision kind. The paper makes formal and operational commitments only. It claims no empirical validation and no universal optimisation theorem.

1 Scope: a decision grammar, not an implementation

Seldon defines $\mathcal{S}_{h,\rho}(\mathcal{F})$, the structured representation at horizon h and resolution ρ of future histories reachable under represented admissible interventions and boundary conditions (Definition 6.1). It distinguishes a local admissibility condition from non-zero-horizon future geometry (Proposition 6.2), and represents goals, where held, as acceptable subsets (Definition 11.1). None of those constructions selects an action. This paper concerns that missing question: how a frame selects, retains, or reopens a decision over its represented future geometry.

The boundary is strict. This is not a theory of search machinery, neural activity, runners, writes, motor control, or physical execution. An agent may retain a Narrative-based decision where implementation is expensive, delayed, or fails. Those are distinct questions about carrying,

constraining, or realising a decision. An external change that makes no difference to the represented landscape makes no direct difference to this decision grammar.

The paper also does not ask whether the grammar is "true" in a frame-free sense. A proposed grammar is useful only if, in a specified frame class and against a stated baseline, it improves prediction of decision persistence or revision, or changes selection toward a stated target at an acceptable additional inference cost. That is an operational commitment, not an ontological claim.

2 The decision frame

Definition 2.1 (Decision frame)**.** *For a current frame $\mathcal{F}$, let its decision description be*

$$D(\mathcal{F}) = (A_{\mathcal{F}}, L_{\mathcal{F}}, S_{\mathcal{F}}, \Gamma_{\mathcal{F}}, N_{\mathcal{F}}, V_{\mathcal{F}}, R_{\mathcal{F}}),$$

where $A_{\mathcal{F}}$ is aperture, $L_{\mathcal{F}}$ the represented landscape, $S_{\mathcal{F}}$ the standing Narrative, $\Gamma_{\mathcal{F}}$ represented admissibility, $N_{\mathcal{F}}$ Narrativium, $V_{\mathcal{F}}$ active comparison descriptions, and $R_{\mathcal{F}}$ represented residual.

Aperture and residual are inherited from Graphic Equalisation (Definitions 7.2 and 8.1). In particular, an aperture omission is absent to the frame, rather than automatically a residual. Narrativium is inherited as frame-relative capacity for consequential deformation (Definition 3.1); it is a state measurement and feasibility relation, not expenditure, energy, a conserved balance, or a universal utility.

Definition 2.2 (Decision domain)**.** *The domain of decision evaluation is $L_{\mathcal{F}}$ only. A candidate is not silently assessed as a negative option when it is outside $A_{\mathcal{F}}$ or absent from $L_{\mathcal{F}}$.*

Remark 2.3 (Agent neutrality and coupling). The definition applies to an individual, business, state, institution, or coalition. At collective scale, aperture may be distributed across offices and information systems; Narrative may be carried by institutional memory, procedures, and commitments; admissibility may be carried by authority and rules; and meta-analysis may require coordination. These are differences in representation and cost, not in decision kind.

A collective's partial frames differ by degree and structure of coupling, not by ontology. Shared representation, communication, authority, binding procedure, memory, commitment, and coordinated capacity can make the partial frames more mutually constraining. A so-called group mind is the high-coupling end of that continuum, not a distinct kind of agent. Where coupling is weak, the proper object of analysis is an interaction among partial frames; where it is strong, an effective collective frame can be used at the operative scale.

3 Narrative continuation and decision opening

The momentum paper defines free continuation as what persists where no rule writes against standing history-bearing structure (Definition 5.2). At the decision level, this yields a narrower reading.

Definition 3.1 (Narrative default)**.** *If the continuation induced by $S_{\mathcal{F}}$ remains represented and admissible, retaining it requires zero* additional decision inference*. This says nothing about its implementation, whose cost may be positive, negative, delayed, or prohibitive.*

Definition 3.2 (Decision opening)**.** *A standing decision is reopened only by either a represented surprise $\sigma_{\mathcal{F}}$, or a current-frame meta-analysis $m_{\mathcal{F}}$:*

$$\text{Change}(\mathcal{F}) \implies \text{Surprise}_{\mathcal{F}}(\sigma_{\mathcal{F}}) \vee \text{MetaAnalyse}_{\mathcal{F}}(m_{\mathcal{F}}).$$

Meta-analysis is an additional decision-inference operation. Its cost and the conditions that trigger it vary by frame. It is not implementation cost.

Proposition 3.3 (Direct surprise revision is aperture-resolved)**.** *A surprise directly licenses revision only if it has a resolution x already available in the current aperture and represented landscape, and that resolution is admissible and coherent in the current frame:*

$$\text{SurpriseRevision}_{\mathcal{F}}(\sigma_{\mathcal{F}}, x) \implies x \in L_{\mathcal{F}} \text{ within } A_{\mathcal{F}} \ \wedge \ \Gamma_{\mathcal{F}}(x) \ \wedge \ \text{Cohere}_{\mathcal{F}}(\sigma_{\mathcal{F}}, x).$$

Proof. The first conjunct is required by the decision domain (Definition 2.2); an unrepresented resolution is not yet a candidate. The second is required by admissibility. The third states that a resolution cannot borrow its warrant from a successor frame not yet installed. Without all three, the mismatch can be represented as surprise but not as a directly selectable replacement. □

3.1 Expected non-occurrence is a negative observation

Definition 3.4 (Expected non-occurrence)**.** *Let X be a represented event and Y a represented opportunity condition or recurrence interval. The frame holds a negative observation $\neg X$ over Y only if all of the following hold:*

(i) $\mathcal{F}$ represents an expectation that X occurs under Y, such as a recurrent relation "X each Y";

(ii) Y is represented as having occurred and the relevant observation interval has elapsed;

(iii) the current aperture can distinguish non-occurrence from an undisclosed or unobserved occurrence; and

(iv) X is not represented in that interval.

Only then may $\neg X$ be treated as candidate surprise rather than as silence or missing information.

Proposition 3.5 (Non-occurrence constrains, but does not identify, change)**.** *If an expected non-occurrence satisfies Definition 3.4, it constrains the represented Seldon manifold by making continuations that require X under the observed Y non-coherent with the current observation. It does not identify a unique hidden change.*

Proof. The represented recurrence and observed opportunity make continuations requiring X incompatible with the represented absence. Several distinct represented causes can remain compatible with that absence: a changed boundary condition, a changed intervention, a changed coupling, a changed recurrence, or an observation failure not excluded by aperture. Hence the absence narrows the represented future geometry without determining one causal explanation. □

Remark 3.6 (Counting what did not happen)*.* A repeated event stopping is therefore not a mere lack of data. Given the represented expectation and adequate aperture, it is data of negative form. It may be surprise and thereby open decision. Direct revision still requires the aperture-available, frame-coherent resolution of Proposition 3.3; otherwise the absence may motivate meta-analysis to distinguish the surviving manifold shapes.

Remark 3.7 (Unresolved surprise)*.* An unresolved or frame-incoherent surprise may motivate meta-analysis. That operation may refine aperture, manifold resolution, active criteria, or the relation between them. It does not entail that the Narrative changes; it only pays to reopen comparison.

4 Target shapes and feasible selection

Definition 4.1 (Target shape)**.** *Where $\mathcal{F}$ represents a Seldon manifold, a target is an acceptable shape*

$$\mathcal{T}_{\mathcal{F},h,\rho} \subseteq \mathcal{S}_{h,\rho}(\mathcal{F}),$$

which may contain regions of future states, families of histories, relations, or constraints. It is a represented object in $\mathcal{F}$, not a hidden privileged destination.

Proposition 4.2 (A target shape is not a point with an error bar)**.** *A point estimate $\hat{z} \pm \epsilon$ represents uncertainty about one intended location or value. A target shape represents acceptability: distinct configurations may be acceptable while nearby configurations may not be. The two coincide only if the acceptable set is a small simply connected tolerance region around one point.*

Proof. The first object is indexed by uncertainty around $\hat{z}$. The second is membership in $\mathcal{T}$. A disconnected, non-convex, or constraint-defined $\mathcal{T}$ cannot in general be recovered from a point and one scalar radius. □

Definition 4.3 (Eligible decision set)**.** *Let*

$$\mathcal{E}_{\mathcal{F}} = \{x \in L_{\mathcal{F}} \mid \Gamma_{\mathcal{F}}(x) \ \wedge \ \textit{any applicable Narrativium feasibility gate holds}\}\,.$$

A transition may be desirable yet absent from $\mathcal{E}_{\mathcal{F}}$ because its represented consequential deformation is not currently reachable.

Proposition 4.4 (Pareto comparison does not select)**.** *Let $v_{\mathcal{F}} : \mathcal{E}_{\mathcal{F}} \to \mathbb{R}^k$ be declared comparison dimensions and let $\mathcal{P}_{\mathcal{F}}$ be its non-dominated subset. Membership in $\mathcal{P}_{\mathcal{F}}$ does not select a unique candidate.*

Proof. Two candidates can be mutually non-dominated. Then Pareto comparison supplies no ordering between them. A priority order, scalarisation, lexicographic rule, or another stated selection relation is still required. □

After a decision opening, selection can ask whether an eligible transition reaches or remains within $\mathcal{T}_{\mathcal{F},h,\rho}$, then apply its stated selection relation among qualifying histories. Narrativium is still a capacity gate, not a value of one target point over another. Absent surprise and absent meta-analysis, the standing Narrative remains selected regardless of visible non-dominated alternatives.

5 Self-decision

Proposition 5.1 (Successor landscapes are present candidates)**.** *A currently represented transformation τ may have a successor decision frame*

$$\mathcal{F}' = \tau(\mathcal{F}),$$

with changed aperture, target shape, Narrative, admissibility, or future manifold. The current frame evaluates τ only through what it represents and orders now.

Proof. Admissibility conditions are functions of antecedent state by the antecedent-state principle (Proposition 5.2). The newly installed frame cannot authorise the transition that first installs it (Corollary 5.3). Therefore a successor landscape is a current candidate, not a prior authority. □

Remark 5.2 (No homunculus and no future veto)*.* This is a structural constraint, not a claim that every human or artificial system actually makes self-revisions in this way. It permits direct self-change under a present frame's rules and capacity, but prevents a successor-only value, identity, or order from supplying the initial warrant.

6 Decision quality, implementation, and outcome

The grammar is not an ideal-decision theory. A normal decision retains Narrative without opening. A frame-valid but poor decision can cohere within a limited or miscalibrated aperture. A well-considered decision may pay for meta-analysis and still fail. Implementation may fail while the decision remains unchanged, and a decision may be implemented as selected yet lead to undesirable realised consequences. These are distinct diagnosis classes.

This distinction keeps outcome learning correctly typed. A later consequence may become represented surprise, which can then have an aperture-available, frame-coherent resolution or motivate meta-analysis. It does not make the external consequence a retroactive input to the original decision.

7 Operational commitments and research neighbours

The operational commitment is comparative. For a stated population, frame class, target shape, horizon, baseline, and inference budget, the grammar must be tested for:

(1) predictive value for persistence, surprise, meta-analysis, revision, and decision-relevant consequences;

(2) decision value for target-shape membership or other declared objectives;

(3) cost of representation, review, and meta-analysis; and

(4) transfer limits across aperture, coupling, and horizon.

A failure to improve these relative to a baseline is a failure of operational usefulness in that scope, not a metaphysical refutation.

Several literatures are close but non-equivalent. Resource-rational metareasoning treats additional reasoning as costly and models selection of reasoning processes (Godara et al., 2024); it does not define Narrative or this opening condition. Default research establishes heterogeneous effects of preselected options (Jachimowicz et al., 2019); it does not establish a general history-bearing Narrative law. Active inference supplies a Bayesian policy model using prediction and preference models (da Costa et al., 2024); GE residual is not thereby variational free energy, prediction error, or Bayesian surprise. Multiobjective optimisation confirms that Pareto-optimal alternatives require additional preference or selection information for final choice (Pajasmaa et al., 2025). AGM belief revision provides a formal neighbour for retention and change of commitments (Alchourrón et al., 1985); it is not identical to Narrative revision.

8 Limits

The paper does not establish that any nervous system, organisation, country, or artificial agent instantiates the complete grammar. It does not derive an implementation, an interpersonal Narrativium accounting, a universal target shape, a universal meta-analysis trigger, or a unique choice rule. It does not make a representation of another frame into access to that frame's actual interior. At collective scale it does not posit a categorical group mind; it models coupling among partial frames. Its contribution is a proposed agent-neutral decision specification over a represented manifold, answerable only by bounded comparative operational tests.

References

Carlos E. Alchourrón, Peter Gärdenfors, and David Makinson. On the logic of theory change: Partial meet contraction and revision functions. *Journal of Symbolic Logic*, 50(2):510–530, 1985.

Lancelot da Costa, Samuel Tenka, Dominic Zhao, and Noor Sajid. Active inference as a model of agency. *arXiv preprint arXiv:2401.12917*, 2024.

Prakhar Godara, Tilman Diego Alemán, and Angela J. Yu. Metareasoning in uncertain environments: A meta-BAMDP framework. *arXiv preprint arXiv:2408.01253*, 2024.

Jon M. Jachimowicz, Shannon Duncan, Elke U. Weber, and Eric J. Johnson. When and why defaults influence decisions: A meta-analysis of default effects. *Behavioural Public Policy*, 3(2):159–186, 2019.

Juuso Pajasmaa, Kaisa Miettinen, and Johanna Silvennoinen. Group decision making in multiobjective optimization: A systematic literature review. *Group Decision and Negotiation*, 34:329–371, 2025.